



Infrastructure as code is still code,
why don't we treat it that way?

Pieter Vincken

Who is Pieter?



CONTAINER days LONDON

The cycle

```
(.venv) → 5-tests git:(main) x tofu validate  
Success! The configuration is valid.
```

```
(.venv) → 5-tests git:(main) x tofu show -json tfplan | tf-summarize
```

+-----+ CHANGE RESOURCE +-----+	
add	aws_s3_bucket.this
+-----+	
	aws_s3_bucket_policy.this
+-----+	

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

OpenTofu will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

aws_s3_bucket.this: Creating...

Error: creating S3 Bucket (pieters-bucket): operation error S3: CreateBucket, https response error StatusCode: 400, RequestID: 644d7757-04bf-440e-b5d3-5c3c265389d8, HostID: s9lzHYrFp76ZVxRcpX9+5cjAnEH2R0uNkd2BHfIa6UkFVdtjf5mKR3/eTPFvsiP/XV/VLi31234=, api error MalformedXML: The XML you provided was not well-formed or did not validate against our published schema

```
with aws_s3_bucket.this,  
on main.tf line 1, in resource "aws_s3_bucket" "this":  
  1: resource "aws_s3_bucket" "this" {
```

Plan: 2 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

OpenTofu will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

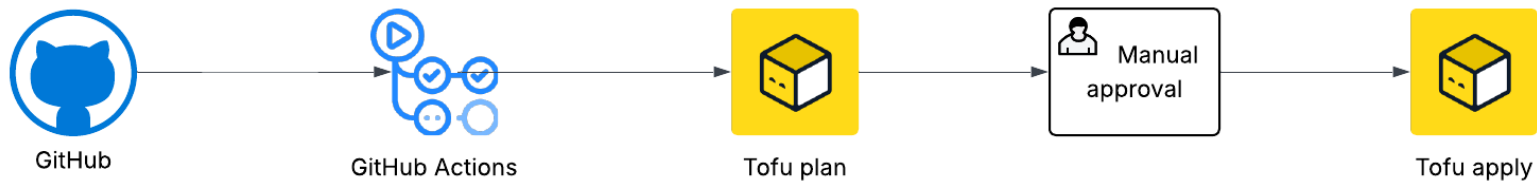
aws_s3_bucket.this: Creating...

Error: creating S3 Bucket (pieters-bucket): operation error createBucket: response error StatusCode: 400, RequestID: 644d7757-04bf-440e-b5d3-5c3c265389d8, HostID: s9lzHYrFp76ZVxRcpX9+5cjAnEH2R0uNkd2BHFia6UKfvutjT5mKR3/eTPFvs134=, api error MalformedXML: The XML you provided was not well-formed or did not validate against our published schema

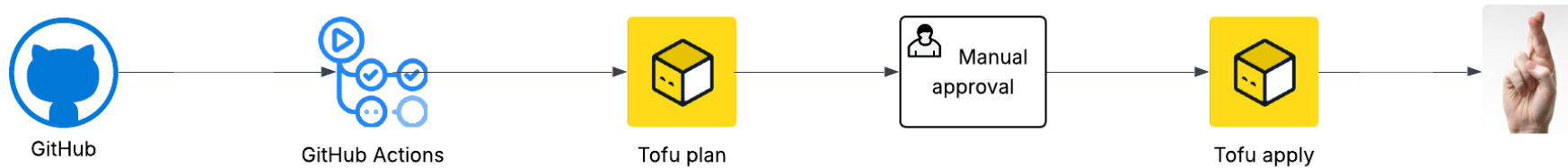
with aws_s3_bucket.this,
on main.tf line 1, in resource "aws_s3_bucket" "this":
1: resource "aws_s3_bucket" "this" {



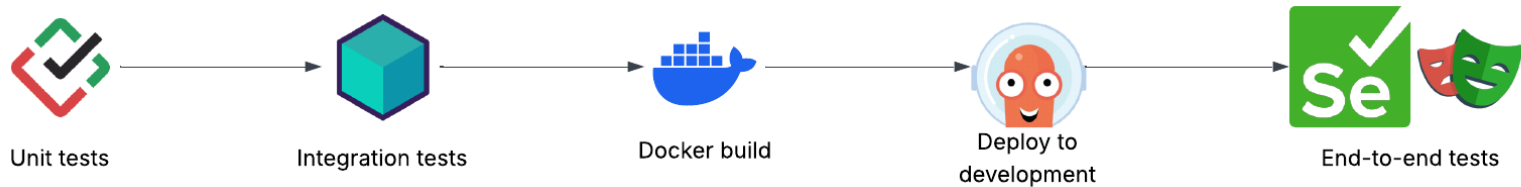
Infrastructure provisioning



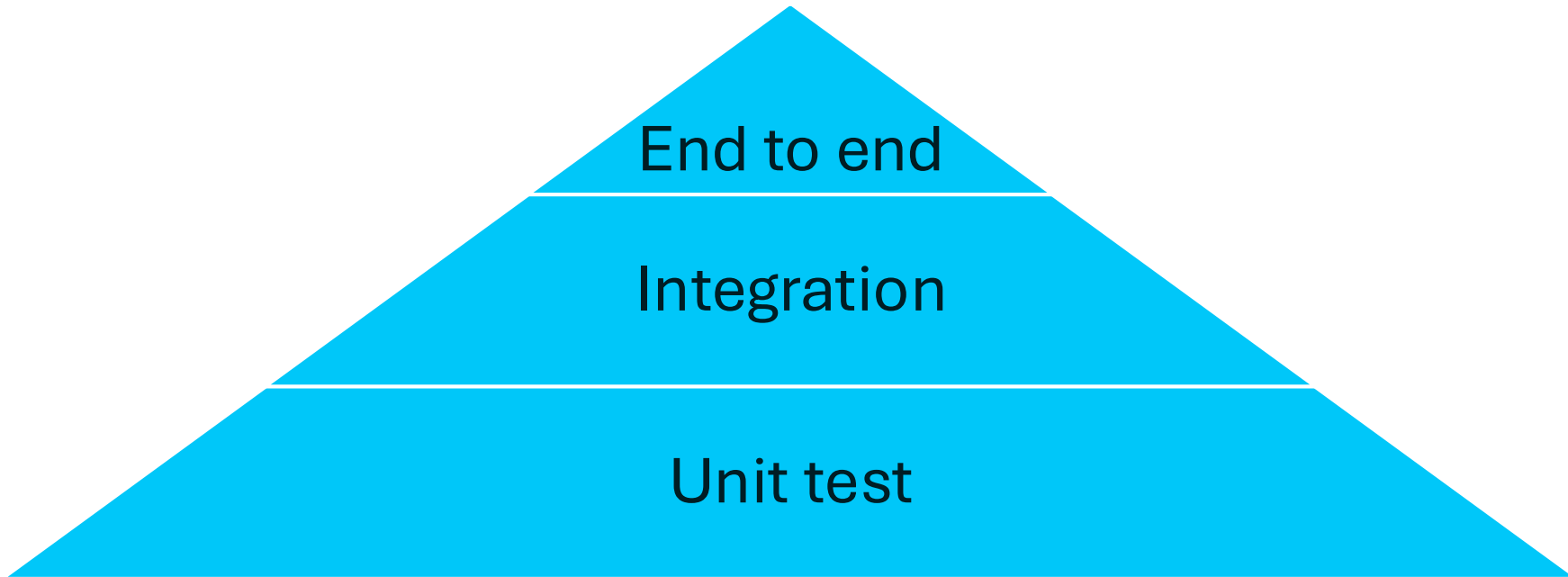
Infrastructure provisioning



Java container builds



Testing pyramid



Terminology

- **Orchestration repository:** Repository with IaC code that is deployable as-is towards an environment.
- **Module repository:** A module is a reusable component. This module should provide a building block solution (combine multiple services and resources). They should be opinionated.

Unit test

Validate inner workings of module

Given

- Dependencies
- Input variables

When

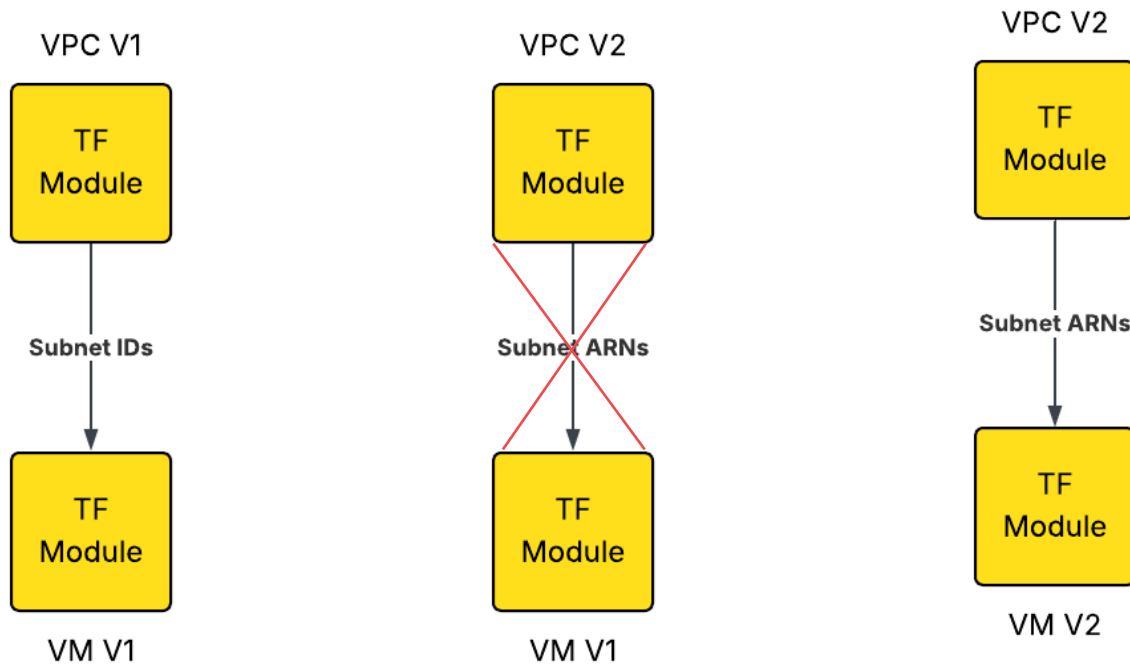
- Plan or apply

Then

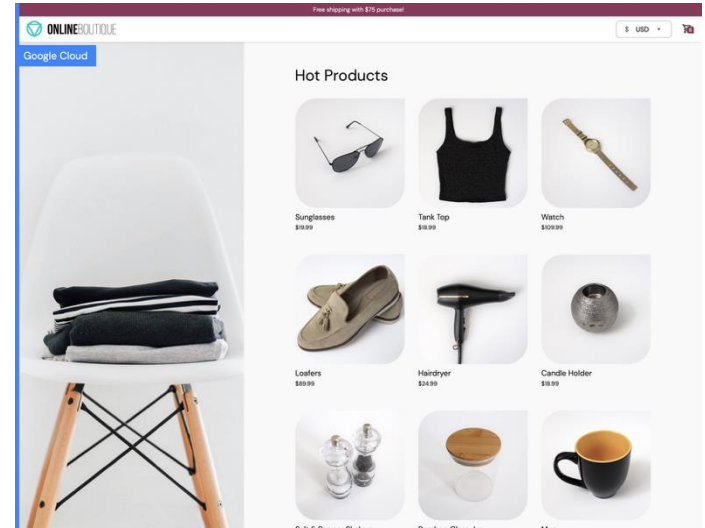
- Output variables
- Deployed configuration

Integration / Contract tests

Validate that module still works with other modules



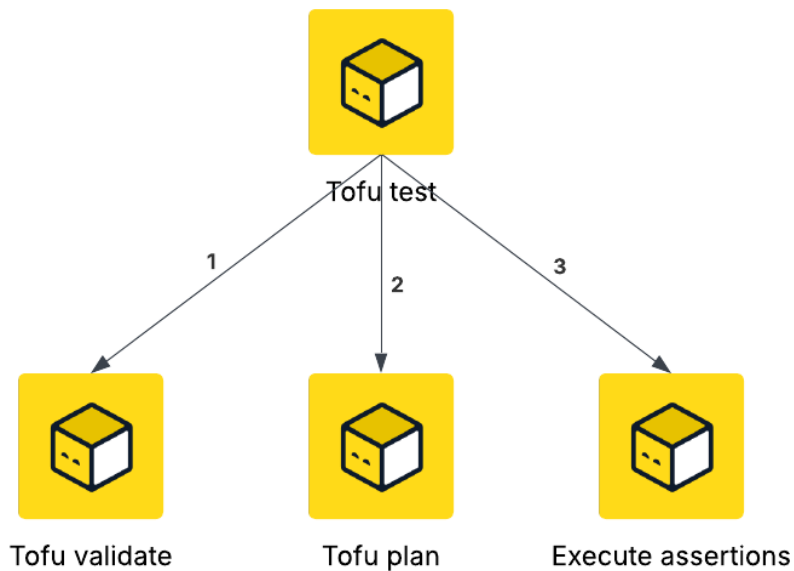
Deployment validation & End-to-end tests



Demo time

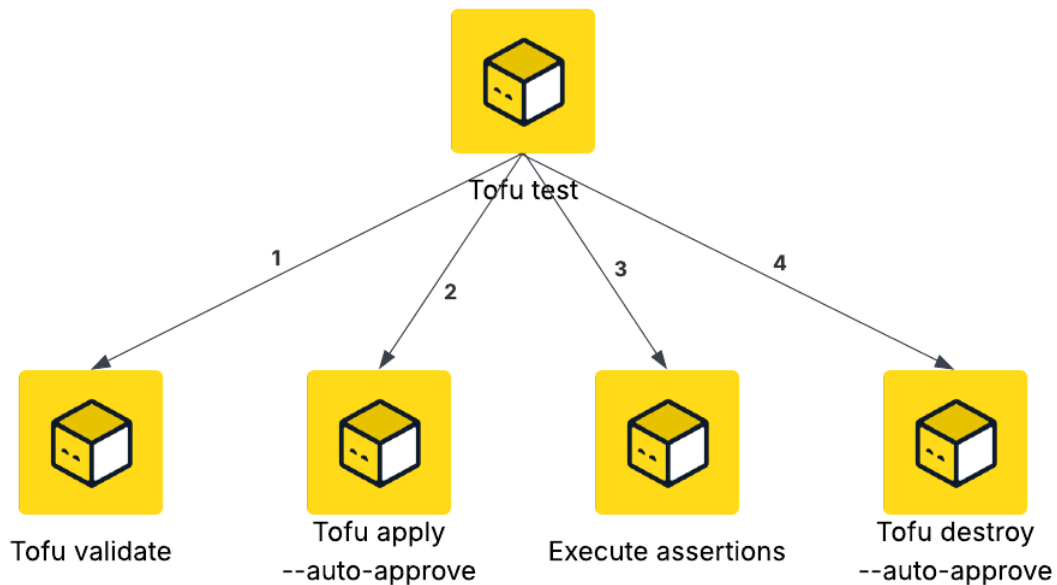
Terraform testing: mode plan

Behind the scenes



Terraform testing: mode apply

Behind the scenes

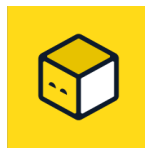


S-tier

Developer laptop



Localstack



tofu test



tofu test
integration-tests



tofu fmt
-recursive

Infrastructure testing tier list

For modules

S	tflocal test tofu fmt -check && tofu validate && tofu plan && tofu test && tofu test integration-test
A	tofu fmt -check && tofu validate && tofu test && tofu test integration-test
B	tofu fmt -check && tofu validate && tofu test
C	tofu fmt -check && tofu validate
D	tofu validate

Infrastructure testing tier list

For orchestration repositories

S	tflocal apply && tflocal test end-to-end-test && tflocal destroy tofu fmt -check && tofu validate && tofu plan -out tfplan && manual review && tofu apply tfplan && tofu test end-to-end-tests
A	tofu fmt -check && tofu validate && tofu plan -out tfplan && manual review && tofu apply tfplan && tofu test end-to-end-tests
B	tofu fmt -check && tofu validate && tofu plan -out tfplan && manual review && tofu apply tfplan
C	tofu plan -out tfplan && manual review && tofu apply tfplan
D	tofu apply -auto-approve



GH repo



Blog



Q&A



Pieter Vincken

Public Cloud Solutions
Architect @Cegeka